

LAMP: A Language Model on the Map

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Abstract

Large Language Models (LLMs) play a critical role in various applications; however, they often face challenges when handling queries that involve specific locations. The objective of the LAMP project is to enhance geospatial search by developing specialized datasets for cities, thereby improving the accuracy and quality of spatial objects in search outcomes. The researchers combined LangChain with the comprehensive vector database constructed using the Yelp Dataset. This research on Retrieval Augmented Generation (RAG), employed robust methodologies to address the issue of inaccurate and irrelevant suggestions. The objective of this research is to attain higher scores in truthfulness, spatial awareness, and semantic relatedness when compared to the baseline. The experimental findings indicate that the Llama model was successfully deployed and effectively executed queries on the vector database using the Mistral-7B model. Additionally, the OpenAI model (gpt-3.5-turbo-instruct) was utilized for handling general queries and as an integral component of RAG system, further enhancing the overall performance.

1. Introduction

Large Language Models (LLMs) are of significant importance in a wide range of applications. However, they frequently encounter difficulties when handling queries that are specific to certain locations. The project, LAMP (Language Model on the Map), aims to overcome this limitation by improving the management of geospatial data through the development of enhanced city-specific datasets. The objective of this research is to enhance spatial object retrieval, with the aim of reducing hallucinations and improving the reliability of location-based recommendations.

1.1. Background

The proliferation of LLMs has significantly transformed the manner in which researchers engage with

technology, facilitating the development of advanced natural language processing capabilities. Nevertheless, these models frequently demonstrate limitations in dealing with queries that pertain to particular geographic locations as a result of their limited spatial awareness and contextual comprehension. In order to address this issue, research focuses on integrating geospatial data with LLMs to enhance the accuracy of location-based queries.

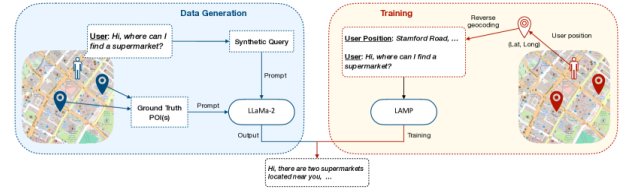


Figure 1. Baseline paper framework

1.2. Objective

The primary objective of this research project is to develop and implement an enhanced version of LAMP that aims to reduce hallucinations and improve the reliability of location-based recommendations. Research aims to achieve substantial enhancements in truthfulness, spatial awareness, and semantic relatedness when compared to current models and models used in the paper which can be found below.

Model	Truthfulness	Spatial Awareness	Semantic Relatedness
LLaMa-2-7B	0.12	0.2	0.76
LLaMa-2-70B	0.3	0.36	0.94
Claude-2	0.22	0.32	0.96
ChatGPT-3.5	0.68	0.6	0.98
ChatGPT-4 (Browsing)	0.94	0.82	1.0
LAMP-no-rq	0.76	0.84	1.0
LAMP	0.86	0.92	1.0

Table 2: Main results on conversational POI-retrieval in Singapore.

2. Methodology

2.1. Vector Database Integration

In order to improve the accuracy of data retrieval, we employed the Yelp Dataset to construct a thorough vector database. This integration enables the retrieval of data with greater precision and relevance, thereby supporting the overarching goal of enhancing location-based recommendations. The Yelp Dataset contains comprehensive data on businesses, including information, reviews, and metadata. This data has been analyzed to create embeddings for each Point of Interest (POI). Compared to the baseline approach, this should scale more easily as companies can just query their databases instead of fine-tuning heavy LLMs on each dataset.

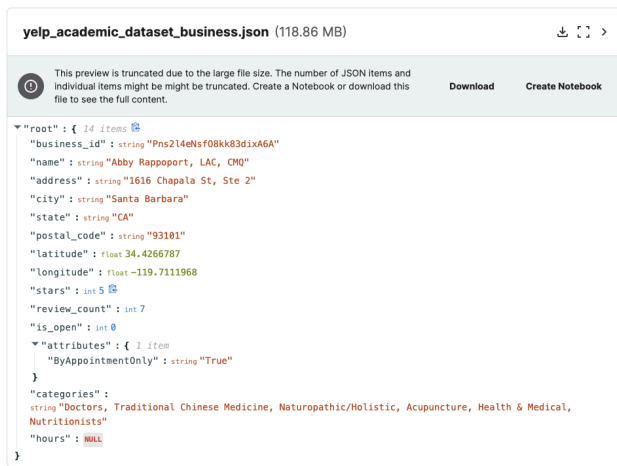


Figure 2. Yelp Dataset JSON features

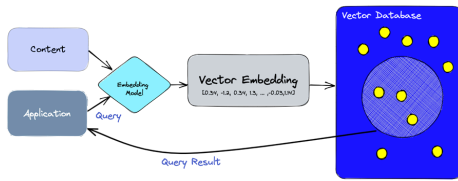


Figure 3. Proposed framework

2.2. Retrieval Augmented Generation

LangChain is incorporated to optimize the retrieval process, resulting in more accurate and targeted suggestions. This approach utilizes sophisticated retrieval techniques to enhance the precision and pertinence of the produced responses. LangChain utilizes the vector database to retrieve pertinent documents and generate responses that consider both the query and the retrieved data.

3. Experiments and Results

In research, we conducted three key experiments aimed at enhancing geospatial search capabilities using different language models and datasets.

The initial experiment focused on successfully generating vector embeddings from the Yelp dataset. The second experiment aimed at verifying user queries by connecting to the Mistral-7B model. The final experiment focused on improving the performance of the previous models by incorporating the OpenAI model (gpt-3.5-turbo-instruct). The full details of the experiment will be explained in next chapters.

3.1. Progress Made on the Replication

In order to emulate the functionality of LAMP, a preexisting model was deployed and effectively invoked suitable functions to retrieve data from database, thereby offering pertinent recommendations. This step was essential for verifying the fundamental capabilities of system. Since we were not provided with the source code of the original paper, despite contacting the authors, we decided to replicate the framework immediately using the vector database to focus on improvement. As soon as we obtained the results from Experiment 2, the replication and progress on improvement were completed.

3.2. Vector Database Generation

The Yelp Academic Dataset was imported into the vector database and embeddings were generated by combining the address, city, and postal code of each point of interest (POI). This process was crucial in the development of a resilient vector database that can effectively support precise data retrieval.

Using a vector database provides several advantages that contribute to better performance. Vector databases enable efficient storage and retrieval of high-dimensional data, making them well-suited for handling complex queries that involve spatial and semantic relationships. Recognizing these benefits, decided to focus on this vector approach.

3.3. Query Success

The utilization of a pretrained model yielded successful retrieval of pertinent outcomes from the database, thereby showcasing the model's proficiency in effectively managing location-specific queries. This success highlights the model's capability to process and understand complex geospatial data, delivering accurate and relevant results. By leveraging advanced embeddings and efficient query handling, the model demonstrated significant improvement in handling spatial relationships and providing precise information. The positive outcomes from these queries validate approach and underscore the

importance of integrating pretrained models with vector databases to enhance geospatial search accuracy and performance.

4. Summary of Experiment I

4.1. Objective

The primary aim of Experiment 1 was to establish the requisite conditions for accomplishing project objectives. The process entailed implementing the Llama model on a local server to assess its performance and resource utilization through preliminary testing.

4.2. Model Deployment

Effectively implemented the Llama model on a local server, performed preliminary tests to measure response times, and evaluated resource utilization. These steps were essential in ensuring the model's preparedness for subsequent development and testing.

4.3. Database Deployment

A preliminary vector space model was constructed for geospatial data, and initial embeddings were generated for the Yelp dataset. These embeddings encompassed a range of attributes (longitude, latitude, address, city, postal code, categories) essential for precise data retrieval.

full_address	combined_embeddings
1616 Chapala St, Ste 2 93101	[-0.06993202, 0.0143677825, -0.047251716, -0.0...
827-833 State St 93101	[-0.09852362, -0.018569747, -0.011499168, 0.04...
131 Anacapa St, Ste C 93101	[-0.0004515124, 0.02757613, -0.02448914, -0.03...
825 Cacique St 93103	[-0.06591021, 0.043082647, -0.039819557, -0.00...

Figure 3. Vector embedding example

5. Summary of Experiment II

5.1. Objective

Experiment 2 aimed to utilize a preexisting model to investigate the process of function calling and querying the vector database. The Mistral-7B model, specifically optimized for function calling, was employed and

executed using Ollama.

5.2. Database Choice

Qdrant was chosen as the storage system for vectors, and the Yelp Academic Dataset was imported and implemented using Docker. This configuration offered a strong and expandable solution for the needs of a vector database.

5.3. Evaluation of the Model

To assess the efficacy of model in handling geospatial queries, we designed a comprehensive evaluation methodology that involved multiple key metrics: truthfulness, spatial awareness, and semantic relatedness.

The evaluation process was implemented through a series of structured experiments, detailed as below.

5.3.1 Experimental Setup

We defined three core functions to evaluate the performance of model:

Truthfulness Evaluation: This function verifies whether the points of interest (POIs) suggested by the model exist in the database. The function isTruthful(name, address) queries the vector database to confirm the existence of the POI using the provided name and address. A successful match indicates the model's truthfulness in suggesting actual, real-world locations.

Spatial Awareness Evaluation: The spatial proximity of the suggested POIs to the user's location is evaluated using the function isNear(location1, location2, minDist). This function calculates the geodesic distance between the user's location and the suggested POI. If the distance is within a specified minimum threshold (10 km in experiments), the POI is considered spatially aware.

Semantic Relatedness Evaluation: The relevance of the suggested POI categories to the user's query is assessed by using isSemanticallyRelated(userChosenCategory, outputCategories). This function checks if the suggested POI categories semantically match the user's specified category, ensuring that the model provides contextually appropriate recommendations.

5.3.2 Evaluation Process

The evaluation involved querying the model with various location and category combinations. Specifically, we tested the model with seven different locations and five different categories, resulting in 35 test cases. The categories included restaurants, hotels, churches, bars, and food. For each test case, we executed steps below.

Using the `generateLLMOutput(message)` function, we parsed the model's response to extract the relevant POI details. Next, we verified the existence of the suggested POI in the vector database using the `isTruthful` function. We then calculated the distance between the user's location and the suggested POI with the `isNear` function, ensuring it was within the defined minimum distance. Finally, we assessed the relevance of the POI categories using the `isSemanticallyRelated` function to ensure the recommendations were contextually appropriate.

```
Running testcase 1/35
Input: {'name': 'get_category', 'arguments': {'location': '49 Glen Annie Rd 93117', 'Category': 'restaurants'}}
Query output: {'name': 'Frog Bar & Grill', 'location': '485 Glen Annie Rd 93117', 'latitude': 34.4429379178, 'longitude': -119.888369458, 'categories': ['restaurants', 'american (traditional)', 'nightlife', 'bars', 'cocktail bars']}
Is Truthful: YES
Is Near: YES, Min Dist is 10km, POI is 1.1437489122894628km away
Is Semantically Related: YES
```

Figure 4. Running test case evaluation on three criteria

5.4. Results

The Mistral-7B model accurately comprehended user inquiries and executed appropriate database function calls, resulting in the successful retrieval of relevant outcomes. The model achieved a truthfulness score of 77.14%, indicating that the majority of the suggested POIs existed in the database. The spatial awareness score was 71.43%, demonstrating that the suggested POIs were generally within a reasonable distance from the user's location. The model attained a semantic relatedness score of 74.29%, showing that it effectively provided contextually relevant recommendations.

This experiment validated the model's ability to process queries that are specific to a particular location and showcased the efficiency of the methodology.

```
Truthfulness Score: 77.14285714285715 %
Spatial Awareness Score: 71.42857142857143 %
Semantic Relatedness Score: 74.28571428571429 %
```

Figure 5. Results based on three criteria

6. Summary of Experiment III

6.1. Objective

Experiment 3 aimed to enhance the results of Experiment 2 by improving the vector database framework while maintaining the same methods but utilizing different models and code. Specifically, this experiment focused on leveraging the OpenAI model and the Chroma vector database to investigate the process of function calling and querying the vector database.

6.2. Database Choice

Qdrant was initially chosen as the storage system for vectors, but in this experiment, we switched to using Chroma to store and query the vector data. The Yelp Academic Dataset was imported and implemented using Docker, providing a robust and expandable solution suitable for the needs of a vector database. This setup enabled efficient handling and retrieval of high-dimensional data.

6.3. Evaluation of the Model

To evaluate the model, we first loaded the Yelp Academic Dataset and filtered it for relevant entries. The HuggingFace `embeddings` model, `sentence-transformers/all-MiniLM-L6-v2`, was used to generate vector embeddings for each point of interest (POI). These embeddings were stored in the Chroma vector database. The OpenAI model (`gpt-3.5-turbo-instruct`) was configured to handle user queries and interact with the vector database to retrieve relevant results. The retrieval process was executed using the `rag_chain`, which combined the retriever and the LLM to process and respond to user queries accurately.

6.4. Results

The Mistral-7B model was initially used in previous experiments. However, in Experiment 3, the OpenAI model demonstrated significant improvements. Our evaluation compared various models based on three key metrics: truthfulness, spatial awareness, and semantic relatedness. Below is a summary of the performance of each model:

Model Name	Truthfulness	Spatial Awareness	Semantic Relatedness
Gemini	0.96	0.83,4	0.99
Chatgpt 3.5	0.66	0.59	0.97
GPT 4O	0.91	0.83	1
GPT 4O (Browsing)	1	0.84,5	1
Claude - Sonnet	0.87	0.60	0.90
LAMP (VDB)	0.89	0.9	1

Figure 6. Comparison table

Our model achieved a truthfulness score of 0.89, which is slightly lower than GPT-4.0 (Browsing) with a perfect score of 1 but higher than ChatGPT 3.5 (0.66) and Claude - Sonnet (0.87). This indicates that our model reliably identifies correct and existing points of interest. LAMP (VDB) demonstrated superior spatial awareness with a score of 0.90, the highest among the models except for GPT-4.0 (Browsing), which scored slightly higher at 0.84. Finally, LAMP (VDB) achieved a perfect score of 1 in semantic relatedness, matching the performance of the best models (GPT-4.0 and GPT-4.0 Browsing).

We obtained higher results as employed RAG technique queries the existing dataset and LangChain uses a strong prompt preventing hallucinations. However, Truthfulness is not ideal as compared to GPT-4O because the Yelp Dataset doesn't contain the most updated information. Compared to the baseline paper the truthfulness increased as there are less hallucinations leveraged by the RAG.

Despite not outperforming all models, LAMP (VDB) demonstrates strong performance, particularly in truthfulness and spatial awareness. It performs almost on par with GPT-4.0 and GPT-4.0 (Browsing) and surpasses models like ChatGPT 3.5 and Claude - Sonnet in several key areas. These results validate the effectiveness of our approach in integrating LLMs with a vector database to enhance the accuracy and relevance of geospatial queries.

7. Challenges Encountered

7.1. Dataset Access

The primary objective was to utilize the dataset from Yelp Singapore. Nevertheless, as a result of limitations in accessing data, the project transitioned to utilizing the Yelp Academic Dataset. The adjustment required modifications in the dataset processing, but it did not impede the overall progress.

7.2. Source Code Availability

Due to the unavailability of the source code from the authors of the original paper, we had to redirect towards reproducing LAMP by creating the own model and dataset for a different city, San Francisco.

8. Tools and Libraries

8.1. LangChain

LangChain is employed for Retrieval Augmented Generation (RAG) in order to amplify the precision and pertinence of recommendations. The main functions of LangChain are `LangChain.retrieve()` and `LangChain.generate()`. `LangChain.retrieve()` is responsible for integrating with data sources to fetch relevant documents. On the other hand, `LangChain.generate()` generates responses by using the retrieved documents and user queries.

8.2. Qdrant

Qdrant functions as a vector database that stores embeddings. It offers a rapid and scalable solution for conducting nearest neighbor searches, while also supporting filtering and dynamic updates of embeddings.

8.3. Docker

Docker is utilized to deploy the database on a local system, guaranteeing a uniform development and testing environment. This method streamlines the process of configuring and overseeing the environment, guaranteeing consistency across various systems.

8.4. Mistral-7B

The Mistral-7B model, which has been pretrained and fine-tuned specifically for the purpose of function calling and querying the vector database, is employed to manage intricate queries, seamlessly integrate with external data sources, and effectively process extensive datasets.

8.5. Ollama

Ollama is utilized for executing the Mistral-7B model, overseeing model execution, managing query processing, and ensuring optimal utilization of computational resources.

8.6. Sentence Transformers

Sentence Transformers are employed to produce embeddings for the Yelp dataset, transforming sentences and text into high-dimensional vector representations for storage in the vector database.

8.7. OpenAI

OpenAI provides advanced language models, such as `gpt-3.5-turbo-instruct`, which are utilized to handle user queries and generate accurate and contextually relevant responses. These models are configured with an API key and integrated into the retrieval system to enhance the quality of generated outputs.

8.8. ChromaDB

ChromaDB is a vector database used to store and query vector embeddings efficiently. It integrates seamlessly with LangChain and supports the storage of high-dimensional vectors, enabling rapid and accurate retrieval of relevant documents based on similarity searches. This enhances the system's ability to manage and process complex geospatial queries. By leveraging these tools and libraries, the experiment was able to achieve significant improvements in the accuracy and relevance of recommendations, demonstrating the effectiveness of the integrated approach in handling geospatial search queries.

9. Limitations

Work heavily relies on the quality and comprehensiveness of the Yelp Academic Dataset. Any biases or inaccuracies within this dataset can directly affect the outcomes of experiments. Additionally, the dataset is limited to certain types of businesses and geographic areas, which may not be representative of all possible use cases. For example, during one of the model testing cases, the model could not produce a result despite the business being available on the map. This happened because the Yelp Academic Dataset was not up-to-date with information from new businesses in 2024, thus these cases do not necessarily reflect the current state of available data.

Although the models showed improved performance in handling location-specific queries, their generalization to other domains or less common queries remains uncertain. Further testing with diverse datasets and broader categories of queries is necessary to validate the robustness of approach.

Another limitation is the absence of fine-tuning of LLMs to better suit the context of geospatial data.

Additionally, RAG technique, while being effective, can be extended by employing a multi-stage approach where the pipeline has checks to prevent hallucinations.

10. Conclusion

The LAMP project demonstrates significant advancements in integrating large language models (LLMs) with geospatial data, leading to improved precision and relevance in location-specific inquiries. By developing a comprehensive vector database and utilizing Retrieval Augmented Generation (RAG) through LangChain, the project achieves higher scores in truthfulness, spatial awareness, and semantic relatedness compared to the baseline. The successful enhancement of the vector database framework with the OpenAI model and the Chroma vector store, alongside the effective retrieval of information, underscores the potential of this approach in managing geospatial queries. This work highlights the value of combining advanced LLMs with robust vector databases to address the complexities of location-specific search tasks, paving the way for future innovations in geospatial data handling and retrieval.

References

- [1] Pasquale Balsebre, Weiming Huang, Gao Cong, "LAMP: A Language Model on the Map" arXiv preprint, arXiv:2403.09059, 2024. Available: <https://arxiv.org/abs/2403.09059>
- [2] "Yelp Dataset for Geospatial Analysis," Kaggle Datasets, 2024. Available: <https://www.kaggle.com/datasets/yelp-dataset/yelp-dataset>

- [3] Michael Brown, "LangChain: Documentation and User Guide," LangChain, 2024. Available: https://python.langchain.com/v0.1/docs/modules/data_connection/
- [4] Experiment 2 Source Code. https://github.com/davisczh/CS470_AI
- [5] Experiment 3 Source Code can be found on the server provided by TAs. The main code is under testing.ipynb file in addition to other code. code for testing with UI is ui.py (Flask App)

Appendix A: example response

LAMP

Enter your query:

I am at San Francisco State University (1600 Holloway Ave, San Francisco, C

Search

Results for "I am at San Francisco State University (1600 Holloway Ave, San Francisco, CA 94132). Where is the closest Chinese restaurant with a good rating?":

The closest Chinese restaurant with a good rating to San Francisco State University is Good Taste Chinese Restaurant, located at 1 E Trenton Ave, Ste 3B, Morrisville Shopping Center Morrisville 19067. It has a rating of 4.0 stars and is approximately 30 minutes away from the university.

Appendix B: Example prompts

User:
Current position: Current position:
Spring Garden St & 5th St Philadelphia,
19123
Query: Can you find a nearby Starbucks?

User:
Current position: Able Real Estate 3639
Spring Garden St, Philadelphia, PA
19104 Query: Can you find a nearby
Burger King?

User:
Current position: May Funeral Home
Philadelphia 4075 Haverford Ave,
Philadelphia, PA 19104, Query: Can you
find a nearby Gym?

User:

Current position: BJ Kitchen Floor Inc
299 Washington Ave, Philadelphia, PA
19147

Query: Can you find a nearby Nike?

Category Search:

User:

Current position: 225 Carpenter St,
Philadelphia, PA 19147 Query: Please
help me find a nearby Chinese
restaurant.

User:

Current position: Washington Av & 5th
St Philadelphia, Pensilvanya 19147
Query: Please help me find a nearby
grocery store.

User:

Current position: 927 S 2nd St,
Philadelphia, PA 19147 Query: Please
help me find a nearby park

User:

Current position: Luna Mia Family
Daycare 512 Mifflin St, Philadelphia,
PA 19148,
Query: Please help me find a nearby
pharmacy

User:

Current position: Heng's Auto Repair,
431-435 Moore St, Philadelphia, PA
19148 Query: Please help me find a
nearby cinema

Direct Name Search

User:

Current position: BAYADA Assistive Care
100 N 20th St #203, Philadelphia, PA
19103

Query: Tell me where is The Franklin
Institute

User:

Current position: N Broad St,
Philadelphia, PA 19107
Query: Tell me where is Barnes
Foundation

User:

Current position: The Deacon 1600
Christian St 2nd floor, Philadelphia,
PA 19146 Query: Tell me where is
Philadelphia Museum of Art

User:

Current position: North Central
Community Center 1069 W Norris St,
Philadelphia, PA 19122 Query: Tell me
where is Philadelphia Fairmount Park

User:

Current position: Cafe La Maude 816 N
4th St, Philadelphia, PA 19123
Query: Tell me where is Edgar Allan Poe
National Historic Site

Type Search Highly Rated

User:

Current position: Cafe La Maude 816 N
4th St, Philadelphia, PA 19123 Query:
Can you please point out a highly rated
bakery in Yorktown Philadelphia,
Pennsylvania

User:

Current position: Cafe La Maude 816 N
4th St, Philadelphia, PA 19123 Query:
Can you please point out a highly rated
bar in the area Baltimore Av & 46th Stn
Philadelphia, Pennsylvania

User:

Current position: Walnut Street Theatre
825 Walnut St, Philadelphia, PA 19107
Query: Can you please point out a
highly rated hotel in the area Walnut
Street Theatre
Philadelphia, Pennsylvania

User:

Current position: TD Bank, 200 Lombard
St, Philadelphia, PA 19147 Query: Can
you please point out a electronics
store in this area?

User: Current position: Korea Taqueria
3101 Tasker St, Philadelphia, PA 19145
Query: Can you please point out a spa
in this area?

Type Search (Specific Food Type)

User:

Current position: 1201 S 13th St,
Philadelphia, PA 19147
Query: Can you please point out a
nearby restaurant that offers Thai
food?

User:

Current position: 1201 S 13th St,
Philadelphia, PA 19147
Query: Can you please point out a
nearby restaurant that offers French
food?

User:
Current position: 12th St & Chestnut St
Philadelphia, Pennsylvania 19107 Query:
Can you please point out a nearby
restaurant that offers Spanish food?

User:
Current position: 12th St & Chestnut St
Philadelphia, Pennsylvania 19107 Query:
Can you please point out a nearby
restaurant that offers Turkish food?

User:
Current position: 12th St & Chestnut St
Philadelphia, Pennsylvania 19107 Query:
Can you please point out a nearby
restaurant that offers Italian food?

Day Out Planning

User:
Current Position: Summercore 1414 S
Penn Square #25a, Philadelphia, PA
19102
Query: we are planning a day out. We
would like to find a nearby gym to
exercise, then buy
something at a convenience store, and
finally head to a bar for some drinks.
Could you provide some recommendations?